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USN

Question Bank: CIE and SEE papers, 2019-2026

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	Compiled from past papers	**Maximum Marks**	–
Course Code	IM352IA	**Duration**	–
Sem	V	**CIE**	Question Bank
OPERATIONS MANAGEMENT			

Note:

1. Source paper and unit are shown under every question. Part A: fill-in-the-blanks and 2-mark questions. Part B: 6 to 20 mark descriptive questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Operations ensure the efficient use of _____ to meet customer demands. *Unit I | CIE-I, 25/11/2024*	01	1	1
2	A supply chain includes the flow of materials, information, and _____ across organizations. *Unit I | CIE-I, 25/11/2024*	01	1	1
3	Effective supply chain management ensures timely delivery and reduces _____. *Unit I | CIE-I, 25/11/2024*	01	1	1
4	Competitive priorities like cost, quality, and _____ determine how a company competes in the market. *Unit I | CIE-I, 25/11/2024*	01	1	1
5	The Fourth Industrial Revolution integrates technologies like IoT, _____, and robotics into operations. *Unit I | CIE-I, 25/11/2024*	01	1	1
6	Processes can be categorized as _____ or supporting based on their direct impact on the final product. *Unit I | CIE-I, 25/11/2024*	01	1	1
7	Differentiate between order winner and order qualifier. *Unit I | CIE-I, 25/11/2024*	02	2	2
8	A hospital is considering a new procedure to be offered at Rs. 200 per patient. The fixed cost per year would be Rs. 100,000, with total variable costs of Rs. 100 per patient. What is the break-even quantity for this service? *Unit I | CIE-I, 25/11/2024*	02	2	2
9	When Amazon decides where to locate its fulfillment centers, it is making a _____ level decision in operations management. *Unit I | CIE-I, 06/11/2025*	01	2	2
10	Decision-making models in operations include deterministic, probabilistic, and _____ models. *Unit I | CIE-I, 06/11/2025*	01	1	3

Sl. No	Questions	M	BT	CO
11	The Fourth Industrial Revolution is driven by technologies such as IoT, AI, and _____. *Unit I | CIE-I, 06/11/2025*	01	1	1
12	When a factory balances machine utilization and customer delivery times, it is managing the _____ responsiveness trade-off. *Unit I | CIE-I, 06/11/2025*	01	2	2
13	Walmart's use of RFID and data analytics to track inventory is an example of using _____ as a driver of supply chain performance. *Unit I | CIE-I, 06/11/2025*	01	2	3
14	Zara's ability to rapidly design and distribute new fashions demonstrates the competitive priority of _____. *Unit I | CIE-I, 06/11/2025*	01	2	2
15	Strategies for change include process improvement, process redesign, and _____ process. *Unit II | CIE-I, 06/11/2025*	01	1	1
16	Strategic fit refers to the alignment between business strategy and _____ strategy. *Unit II | CIE-I, 06/11/2025*	01	2	2
17	In manufacturing, the four process structures are job process, batch process, line process, and _____ process. *Unit II | CIE-I, 06/11/2025*	01	1	1
18	In a make-to-order process, production begins only after receiving a _____ from the customer. *Unit II | CIE-I, 06/11/2025*	01	1	1
19	In services, the process structure is primarily influenced by the level of _____ and customization. *Unit II | CIE-II, 07/01/2025*	01	1	2
20	In manufacturing, a _____ process is ideal for high-volume, standardized production. *Unit II | CIE-II, 07/01/2025*	01	2	2
21	The first step in a systematic approach to long-term capacity decisions is _____ future demand. *Unit II | CIE-II, 07/01/2025*	01	2	2
22	A systematic approach to capacity planning involves forecasting, identifying alternatives, and evaluating _____. *Unit II | CIE-II, 07/01/2025*	01	1	2
23	_____ ensures that operational capabilities are in line with the company's strategic goals. *Unit II | CIE-II, 07/01/2025*	01	2	3
24	A _____ shop deals with customized, low-volume production. *Unit II | Improvement CIE, 12/01/2026*	01	1	2
25	In mass production, the layout used is _____ layout. *Unit II | Improvement CIE, 12/01/2026*	01	1	2
26	The maximum output a system can produce is called its _____. *Unit II | Improvement CIE, 12/01/2026*	01	1	2
27	The long-term decision of choosing a geographic site for operations is called _____. *Unit II | Improvement CIE, 12/01/2026*	01	2	3
28	Continuous processes are suitable for _____ production. *Unit II | Improvement CIE, 12/01/2026*	01	2	3
	Part – B			
1	Define operations management. Briefly explain the key terms involved in the definition of operations management with a schematic model of a production system. *Unit I | CIE-I, 06/11/2025 and 25/11/2024 (same question)*	10	2	1
2	With suitable example, distinguish between manufacturing and service processes. *Unit I | CIE-I, 06/11/2025 and 25/11/2024 (same question)*	10	2	1

<table>
<tr><th>Sl. No</th><th>Questions</th><th>M</th><th>BT</th><th>CO</th></tr>
<tr><td>3</td><td>A retailer must decide whether to build a small or a large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6, respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If a small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7, respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative – building a small facility or building a large facility – has the higher expected payoff?
Unit I | CIE-I, 06/11/2025</td><td>16</td><td>4</td><td>3</td></tr>
<tr><td>4</td><td>A manager is deciding whether to build a small or a large facility. Much depends on the future demand that the facility must serve, and demand may be small or large. The manager knows with certainty the payoffs that will result under each alternative, shown in the payoff table below. What is the best alternative for each decision rule: Maximin, Maximax, Laplace, Minimax Regret?

<table><tr><td>Alternative</td><td>Low demand</td><td>High demand</td></tr><tr><td>Small facility</td><td>200</td><td>270</td></tr><tr><td>Large facility</td><td>160</td><td>800</td></tr><tr><td>Do nothing</td><td>0</td><td>0</td></tr></table>
Unit I | CIE-I, 06/11/2025 (14 M); CIE-I, 25/11/2024 (10 M)</td><td>14</td><td>4</td><td>3</td></tr>
<tr><td>5</td><td>White Valley Ski Resort is planning the ski lift operation for its new ski resort. Management is trying to determine whether one or two lifts will be necessary; each lift can accommodate 250 people per day. Skiing normally occurs in the 14-week period from December to April, during which the lift will operate 7 days per week. The first lift will operate at 90 percent capacity if economic conditions are bad, the probability of which is believed to be about 0.3. During normal times the first lift will be utilized at 100 percent capacity, and the excess crowd will provide 50 percent utilization of the second lift. The probability of normal times is 0.5. Finally, if times are really good, the probability of which is 0.2, the utilization of the second lift will increase to 90 percent. The equivalent annual cost of installing a new lift, recognizing the time value of money and the lift's economic life, is $50,000. The annual cost of installing two lifts is only $90,000 if both are purchased at the same time. If used at all, each lift costs $200,000 to operate, no matter how low or high its utilization rate. Lift tickets cost $20 per customer per day. Should the resort purchase one lift or two?
Unit I | CIE-I, 25/11/2024</td><td>20</td><td>4</td><td>3</td></tr>
<tr><td>6</td><td>Explain the major process strategy decisions and their implications for operations.
Unit II | CIE-II, 07/01/2025</td><td>10</td><td>2</td><td>2</td></tr>
</table>

<table>
<tr><th>Sl. No</th><th>Questions</th><th>M</th><th>BT</th><th>CO</th></tr>
<tr><td>7</td><td>Richard Garber is the head designer for Matthews and Novak Design Company. Garber has been called in to design the layout for a newly constructed office building. From statistical samplings over the past three months, Garber developed the following closeness matrix for daily trips between the department's offices.

<table>
<tr><td>Department</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td></tr>
<tr><td>A</td><td>–</td><td>25</td><td>90</td><td></td><td>105</td><td>185</td></tr>
<tr><td>B</td><td></td><td>–</td><td></td><td></td><td>125</td><td>125</td></tr>
<tr><td>C</td><td></td><td></td><td>–</td><td></td><td>25</td><td></td></tr>
<tr><td>D</td><td></td><td></td><td></td><td>–</td><td></td><td>105</td></tr>
<tr><td>E</td><td></td><td></td><td></td><td></td><td>–</td><td></td></tr>
<tr><td>F</td><td></td><td></td><td></td><td></td><td></td><td>–</td></tr>
</table>

a. If other factors are equal, which two offices should be located closest together?
b. A figure shows an alternative layout for the department: top row C, F, A; bottom row B, E, D. What is the total weighted-distance score for this plan based on rectilinear distance, assuming that offices A and B are three units of distance apart?
Unit II | CIE-II, 07/01/2025</td><td>10</td><td>3</td><td>2</td></tr>
<tr><td>8</td><td>Explain the concepts of economies and diseconomies of scale with appropriate examples.
Unit II | Improvement CIE, 12/01/2026</td><td>10</td><td>2</td><td>2</td></tr>
<tr><td>9</td><td>Explain process reengineering and process improvement, highlighting their differences with suitable examples.
Unit II | Improvement CIE, 12/01/2026</td><td>10</td><td>4</td><td>3</td></tr>
<tr><td>10</td><td>Describe a systematic approach to process analysis with the help of a block diagram.
Unit II | Improvement CIE, 12/01/2026</td><td>10</td><td>2</td><td>2</td></tr>
<tr><td>11</td><td>A copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run, collate, and bind each copy depends on, among other factors, the number of pages. The center operates 250 days per year, with one 8-hour shift. Management believes that a capacity cushion of 15 percent (beyond the allowance built into time standards) is best. It currently has three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

<table>
<tr><td>Item</td><td>Client X</td><td>Client Y</td></tr>
<tr><td>Annual demand forecast (copies)</td><td>2,000</td><td>6,000</td></tr>
<tr><td>Standard processing time (hours/copy)</td><td>0.5</td><td>0.7</td></tr>
<tr><td>Average lot size (copies per report)</td><td>20</td><td>30</td></tr>
<tr><td>Standard setup time (hours)</td><td>0.25</td><td>0.40</td></tr>
</table>

Unit II | Improvement CIE, 12/01/2026</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>12</td><td>Consider amazon.com, whose Website enjoys millions of "hits" each day and puts customers in touch with millions of services and products. What are amazon.com's competitive priorities and what should its operations strategy focus on?
Unit I | SEE Jan/Feb 2021 (18IM54), Q2a</td><td>06</td><td>–</td><td>–</td></tr>
<tr><td>13</td><td>Identify nine competitive priorities used in operations strategy and explain how a consistent pattern of decisions can develop organizational capabilities.
Unit I | SEE Feb/Mar 2025 (IM352IA), Q2a</td><td>10</td><td>2</td><td>1</td></tr>
<tr><td>14</td><td>Describe the process view of operations in typical organizations with the help of a block diagram.
Unit I | SEE Feb/Mar 2025 (IM352IA), Q2b</td><td>06</td><td>2</td><td>1</td></tr>
<tr><td>15</td><td>Differentiate between competitive priorities and competitive capabilities with suitable examples.
Unit I | SEE Apr/May 2024 (21IM53), Q2a</td><td>08</td><td>2</td><td>1</td></tr>
</table>

Sl. No	Questions	M	BT	CO
16	IKEA shows how important operations management is for its own success and the success of any type of organization. In this view identify some of the activities that IKEA's operations managers are involved in through highlighting some eight important aspects. *Unit I | SEE Apr/May 2024 (21IM53), Q2b*	08	3	2
17	Explain the term operations management and identify the three major functional areas of business organizations and briefly describe how they are interrelated. *Unit I | SEE Mar/Apr 2023 (18IM54), Q2a*	08	–	–
18	Describe the new trends and challenges in operations management. *Unit I | SEE Mar/Apr 2023 (18IM54), Q2b*	08	–	–
19	A retailer must decide whether to build a small or large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6 respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If the small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate maximum demand through local advertising. The response to advertising may be either modest or sizable, with probabilities estimated to be 0.3 and 0.7 respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000. Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative – building a small facility or building a large facility – has the higher expected payoff? *Unit I | SEE Jan/Feb 2021 (18IM54), Q2b*	10	–	–
20	Developing a customer-driven operations strategy begins with corporate strategy, which coordinates the firm's overall goals with its core processes. Identify and discuss the factors connecting corporate strategy and key operations management decisions with block diagram. *Unit I | SEE Jan/Feb 2021 (16IM55); Nov/Dec 2019 (16IM55), Q2a (identical)*	08	–	–
21	Calculate the productivity for the following operations: (i) Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week. (ii) A team of workers makes 400 units of a product, which is sold in the market for $10 each. The accounting department reports that for this job the actual costs are $400 for labour, $1,000 for materials, and $300 for overhead. *Unit I | SEE Jan/Feb 2021 (16IM55); Nov/Dec 2019 (16IM55), Q2b (identical)*	08	–	–
22	Define operations management. Explain with the help of a block diagram the process view of operations management and write how processes work. *Unit I | SEE Fast Track Oct 2020 (16IM55/12IM62), Q2a*	06	–	–
23	A manager is trying to decide whether to build a small, medium or large facility. Demand can be low, average or high with estimated probabilities being 0.25, 0.40 and 0.35 respectively. A small facility is expected to earn an after tax net present value of just $18,000 if demand is low. If demand is average, the small facility is expected to earn $75,000; it can be increased to medium size to earn a net present value of $60,000. If demand is high, the small facility is expected to earn $75,000 and can be expanded to medium size to earn $60,000 or to large size to earn $125,000; a medium sized facility is expected to lose an estimated $25,000 if demand is low and earn $140,000 if demand is average. If the demand is high the medium sized facility is expected to earn a net present value of $150,000; it can be expanded to a large size for a net pay off of $145,000. If a large facility is built and demand is high, earnings are expected to be $220,000. If demand is average for the large facility the present value is expected to be $125,000; if demand is low, the facility is expected to lose $60,000. (i) Draw a decision tree for this problem. (ii) What should the management do to achieve the highest expected pay off? *Unit I | SEE Fast Track Oct 2020 (16IM55/12IM62), Q2b*	10	–	–

Sl. No	Questions	M	BT	CO
24	How does the process view and the supply chain view of operations management help us understand the need and importance of operations management as a strategic function in the organization? Elaborate on these two perspectives of operations management. *Unit I | SEE Fast Track Jul 2019 (16IM55/12IM62), Q2a*	08	–	–
25	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before commercializing the device, the owner wants reasonable assurance of success. Variable costs are $7 per unit produced and sold. Fixed costs are about $500,000 per year. (i) If the price is reduced to $15 and forecasted sales for the first year are 10,000 units, what would be the product's total contribution to profits in the first year? (ii) If the selling price is set at $25, how many units must be produced and sold to break even? Use both algebraic and graphical approaches. *Unit I | SEE Fast Track Jul 2019 (16IM55/12IM62), Q2b*	08	–	–
26	The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Variable costs are estimated at $7 per unit produced and sold. Fixed costs are about $56,000 per year. (i) If the selling price is set at $25, how many units must be produced and sold to break even? (ii) Forecasted sales for the first year are 10,000 units if the price is reduced to $15. With this pricing strategy, what would be the product's total contribution to profits in the first year? *Unit I | SEE Feb/Mar 2026 (IM352IA), Q2a*	08	3	1
27	With suitable example, distinguish between manufacturing processes and service processes. *Unit I | SEE Feb/Mar 2026 (IM352IA), Q2b*	08	2	1
28	Compare process structures in services and manufacturing. *Unit II | SEE Feb/Mar 2026 (IM352IA), Q3a*	08	4	3
29	Explain long-term capacity planning. Describe capacity timing strategies and capacity sizing strategies with suitable explanations. *Unit II | SEE Feb/Mar 2026 (IM352IA), Q3b*	08	4	3
30	Explain timing and sizing strategies in capacity planning with diagrams. *Unit II | SEE Feb/Mar 2026 (IM352IA), Q4a*	08	4	3
31	How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer introduced variability? *Unit II | SEE Feb/Mar 2026 (IM352IA), Q4b*	08	2	2
32	Explain the four inventory strategies that should be coordinated with process choice in a manufacturing environment. Provide example for each of the strategy. *Unit II | SEE Feb/Mar 2025 (IM352IA), Q3a*	10	3	2
33	Provide a note on the three elements in a customer-contract matrix in service processes. *Unit II | SEE Feb/Mar 2025 (IM352IA), Q3b*	06	2	2
34	Operations managers must examine three dimensions of capacity strategy before making capacity decisions. What are they? Explain each with suitable examples. *Unit II | SEE Feb/Mar 2025 (IM352IA), Q4a*	10	3	2
35	Enumerate the four steps of systematic approach to long term capacity decisions. *Unit II | SEE Feb/Mar 2025 (IM352IA), Q4b*	06	2	2
36	List and explain operations process characteristics. Also briefly discuss the implication of four characteristics of operations processes. *Unit II | SEE Apr/May 2024 (21IM53), Q3a*	08	3	2
37	How does the degree of customer contact, customization and process characteristics help in structuring processes in a service organization? *Unit II | SEE Apr/May 2024 (21IM53), Q3b*	08	3	2
38	Explain the concept of economies of scale when planning capacity for long term implementation. *Unit II | SEE Apr/May 2024 (21IM53), Q4a*	08	2	2
39	The best choice for a manufacturing process depends on the volume and degree of customization required of the process. Justify the above statement with the help of a suitable matrix. *Unit II | SEE Apr/May 2024 (21IM53), Q4b*	08	3	3

<table>
<tr><th>Sl. No</th><th>Questions</th><th>M</th><th>BT</th><th>CO</th></tr>
<tr><td>40</td><td>Describe the four basic process strategies.
Unit II | SEE Mar/Apr 2023 (18IM54), Q3a</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>41</td><td>List the different tools for process analysis and design and explain them.
Unit II | SEE Mar/Apr 2023 (18IM54), Q3b</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>42</td><td>Define strategic fit. Describe the elements of strategic fit.
Unit II | SEE Mar/Apr 2023 (18IM54), Q4a</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>43</td><td>What is the Six Sigma model of quality improvement? Describe the five steps of the Six Sigma improvement model.
Unit II | SEE Mar/Apr 2023 (18IM54), Q4b</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>44</td><td>How do the process strategies of eBay and McDonald's differ, and how do their choices relate to customer-introduced variability?
Unit II | SEE Jan/Feb 2021 (18IM54), Q3a</td><td>06</td><td>–</td><td>–</td></tr>
<tr><td>45</td><td>A work sampling study was conducted to determine the proportion of the time a worker is idle. The following information was gathered on a random basis.
<table><tr><td>Day</td><td>Number of times worker idle</td><td>Total number of observations</td></tr><tr><td>Monday</td><td>17</td><td>44</td></tr><tr><td>Tuesday</td><td>18</td><td>56</td></tr><tr><td>Wednesday</td><td>14</td><td>48</td></tr><tr><td>Thursday</td><td>16</td><td>60</td></tr></table>(i) Based on these preliminary results, what percent of the time is the worker working? (ii) If the idle time is judged to be excessive, what additional categories might you add to a follow-up work sampling study to identify the root causes?
Unit II | SEE Jan/Feb 2021 (18IM54), Q3b</td><td>10</td><td>–</td><td>–</td></tr>
<tr><td>46</td><td>Performance criteria important in creating a layout can go well beyond communication and materials handling. Identify the types of layout performance criteria that might be most important in the following settings: (i) Airport (ii) Bank (iii) Classroom (iv) Product designers' office.
Unit II | SEE Jan/Feb 2021 (18IM54), Q4a</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>47</td><td>Vera Johnson and Merris Williams manufacture vanishing cream. Their packaging process has four steps: (1) mix, (2) fill, (3) cap, and (4) label. They have had the reported process failures analyzed, which shows the following (60 failures in total): lumps of unmixed product – 7; over or under filled jars – 18; jar lids did not seal – 6; labels rumpled or missing – 29. (No separate instruction is printed in the paper.)
Unit II | SEE Jan/Feb 2021 (18IM54), Q4b</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>48</td><td>Identify and discuss with block diagram the important four decisions that are essential for an effective process design.
Unit II | SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q3a (identical)</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>49</td><td>Identify and discuss the strategies used for structuring the process in manufacturing.
Unit II | SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q3b (identical)</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>50</td><td>The manager should understand how the four major process decisions tie together, so as to spot ways of improving poorly designed processes. The choices should fit the situation and each other. When the fit is more strategic, the process will be more effective. As an industrial engineer you are required to identify, examine and discuss the services and manufacturing processes, looking for ways to test for strategic fit.
Unit II | SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q4a (identical)</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>51</td><td>Identify and discuss the key elements of process reengineering.
Unit II | SEE Jan/Feb 2021 and Nov/Dec 2019 (16IM55), Q4b (identical)</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>52</td><td>Explain why processes exist everywhere in all organizations. Take the following supply chain activities and give a brief description for each: (i) Warehousing (ii) Customer service (iii) Logistics.
Unit II | SEE Fast Track Oct 2020 (16IM55/12IM62), Q3a</td><td>08</td><td>–</td><td>–</td></tr>
<tr><td>53</td><td>Briefly discuss on the process strategy decisions. Use a block diagrammatic representation to bring out the connections among major process strategy decisions for effective processes.
Unit II | SEE Fast Track Oct 2020 (16IM55/12IM62), Q3b</td><td>08</td><td>–</td><td>–</td></tr>
</table>

Sl. No	Questions	M	BT	CO
54	Define the following terms clearly: (i) Customer increment (ii) Resource flexibility (iii) Capital intensity (iv) Economies of scope. *Unit II	SEE Fast Track Oct 2020 (16IM55/12IM62), Q4a*	08	–	–
55	Position a process on the customer-contact matrix or product process matrix. Bring out the characteristics of the following types of process choices: (i) Job processes (ii) Batch processes (iii) Line processes (iv) Continuous flow processes. *Unit II	SEE Fast Track Oct 2020 (16IM55/12IM62), Q4b*	08	–	–
56	Construct the product-process matrix and show in this matrix the process characteristic categories, level of product customization and the elements of volume. Elaborate on the type of manufacturing process structure that exist in various types of organization briefly. *Unit II	SEE Fast Track Jul 2019 (16IM55/12IM62), Q3a*	08	–	–
57	"After a process is documented, metrics data collected and disconnects identified, the process analyst or design team puts together a set of changes that will make the process better". Compare this with a doctor diagnosing the patient and recommending treatments based on the diagnosis. Explain how questioning and brainstorming will assist the process designer in analyzing the gaps in the process. Take an example to support your answer. *Unit II	SEE Fast Track Jul 2019 (16IM55/12IM62), Q3b*	08	–	–
58	Develop the customer-contact matrix and indicate the elements of degree of customer contact, customization and process characteristic. Elaborate on the process divergence and flow based on the customer-contact matrix. How does this matrix provide the starting point for evaluating and improving a process? *Unit II	SEE Fast Track Jul 2019 (16IM55/12IM62), Q4a*	08	–	–
59	Capital intensity is the mix of equipment and human skills in the process. Carefully examine the advantages and disadvantages of automating a manufacturing process. Distinguish between fixed automation and flexible automation. Give two examples for fixed and flexible automation respectively. Justify automation in the context of capital intensity. *Unit II	SEE Fast Track Jul 2019 (16IM55/12IM62), Q4b*	08	–	–